HW7: Review

October 30, 2025

Exercise 1 Consider the following initial value problem for Burgers equation.

$$\partial_t u(t,x) + u(t,x) \cdot \partial_x u(t,x) = 0, \quad u(0,x) = \varphi(x), \quad (t,x) \in [0,\infty) \times \mathbb{R}.$$

1. Let
$$\varphi(x) = \begin{cases} 0, & x \le 0, \\ x, & x > 0. \end{cases}$$
Find a solution $u(t,x) \in \mathcal{C}^{1,1}\big((0,\infty) \times \mathbb{R}\big) \cap \mathcal{C}\big([0,\infty) \times \mathbb{R}\big)$.

2. Let
$$\varphi(x) = \begin{cases} 1, & x \le 0, \\ 1 - \dfrac{1}{2}x, & 0 < x \le 2, \\ 0, & x > 2. \end{cases}$$
Find the largest time t_s such that all characteristics do not intersect.

3. Find the expression of $u(t,x)$ for $t < t_s$.

Exercise 2

1. Find all the eigenvalues and eigenfunctions of the Sturm–Liouville problem:
$$X''(x) + \lambda X(x) = 0, \quad X(0) = 0, \quad X'(\pi) = 0.$$

2. Use Separation of Variables to solve the homogeneous heat equation:
$$\begin{cases} \partial_t u - \partial_{xx} u = 0, & x \in (0,\pi),\ t > 0, \\ u(0,x) = \sin\dfrac{x}{2}, & x \in [0,\pi], \\ u(t,0) = \partial_x u(t,\pi) = 0, & t \ge 0. \end{cases}$$

3. Use Duhamel's principle to solve the inhomogeneous heat equation:
$$\begin{cases} \partial_t u - \partial_{xx} u = f(t,x), & x \in (0,\pi),\ t > 0, \\ u(0,x) = \sin\dfrac{x}{2}, & x \in [0,\pi], \\ u(t,0) = \partial_x u(t,\pi) = 0, & t \ge 0. \end{cases}$$

Exercise 3 Let $\Omega = B_1(0) \subset \mathbb{R}^2$ and $\Omega_T = (0,T] \times \Omega$.

1. Prove the following comparison principle: if $v \in \mathcal{C}^{1,2}(\Omega_T) \cap \mathcal{C}^{0,1}(\overline{\Omega_T})$ satisfies

$$\begin{cases} \partial_t v - \Delta v \geq 0, & (t,x) \in \Omega_T, \\ v(0,x) \geq 0, & x \in \Omega, \\ \dfrac{\partial v}{\partial n}(t,x) > 0, & t > 0,\ x \in \partial\Omega, \end{cases}$$

then $v(t,x) \geq 0$ on $\overline{\Omega_T}$.

2. Assume instead that $\dfrac{\partial}{\partial n} v(t,x) \geq 0$ for all $t > 0$ and $x \in \partial\Omega$. Show that $v(t,x) \geq 0$ on $\overline{\Omega_T}$.

   Hint: consider $v_\varepsilon(t,x) = v(t,x) + \varepsilon(4t + x_1^2 + x_2^2 - 1)$.

3. Let $u \in \mathcal{C}^{1,2}(\Omega_T) \cap \mathcal{C}^{0,1}(\overline{\Omega_T})$ solve

$$\begin{cases} \partial_t u - \Delta u = f(t,x), & (t,x) \in \Omega_T, \\ u(0,x) = \varphi(x), & x \in \Omega, \\ \dfrac{\partial u}{\partial n}(t,x) = 0, & t > 0,\ x \in \partial\Omega. \end{cases}$$

   Show that there exists a constant $C = C(T)$ such that

$$\sup_{\overline{\Omega_T}} |u| \leq C\big(\sup_{\Omega_T} |f| + \sup_{\Omega} |\varphi|\big).$$

Exercise 4 Let $K(x,y) : \mathbb{R}^2_+ \times \mathbb{R} \to \mathbb{R}$ be defined by

$$K(x,y) = \frac{1}{\pi} \frac{x_1}{(x_2 - y)^2 + x_1^2}.$$

1. Show that $\displaystyle\int_{\mathbb{R}} K(x,y)\, dy = 1$, and for every $\varepsilon > 0$ and $z \in \mathbb{R}$,

$$\lim_{h \to 0+} \int_{z-\varepsilon}^{z+\varepsilon} K\big((h,z),y\big)\, dy = 1.$$

2. Let $g : \mathbb{R} \to \mathbb{R}$ be uniformly continuous with $|g| \leq 1$. Let

$$g_h(z) = \int_{\mathbb{R}} K\big((h,z),y\big) g(y)\, dy, \quad h > 0.$$

   Show that $g_h \to g$ uniformly on $\mathbb{R}$.

Ex 1. (1) 特征线为 $\eta(t;x_0)=\begin{cases} x_0 & , x_0\le 0 \\ x_0 t + x_0 & , x_0>0 \end{cases}$

当 $x<0$ 时，$u(t,x)=u(0,x)=0$

当 $x\ge 0$ 时，由 $x_0 t + x_0 = x$ 得 $x_0=\frac{x}{t+1}$，故 $u(t,x)=\frac{x}{t+1}$，即 $u(t,x)=\begin{cases} 0 & , x\le 0 \\ \frac{x}{t+1} & , x>0 \end{cases}$

(2) 设两条特征线 $\varphi(x_1)t+x_1$ 与 $\varphi(x_2)t+x_2$ 在 t_0 相交

即 $\varphi(x_1)t_0+x_1=\varphi(x_2)t_0+x_2$，画图可得 $t_0\ge 2$，故 $t_s=2$

(3) 取 $x=\eta(t;x_0)=\varphi(x_0)t+x_0$，只需对 $t<t_s=2$ 讨论

可得 $x_0=\begin{cases} x-t & , x_0\le 0 \\ \frac{2x-2t}{2-t} & , 0<x_0\le 2 \\ x & , x_0>2 \end{cases}$，代回 $\varphi(x_0)$ 得 $u(t,x)=\begin{cases} 1 & , t\ge x \\ \frac{2-x}{2-t} & , t<x\le 2 \\ 0 & , x\ge 2 \end{cases}$

Ex 2. (1) 通解为 $X(x)=A\sin(\sqrt{\lambda}x)+B\cos(\sqrt{\lambda}x)$

代入边界条件得 $B=0$，$\lambda=(\frac{1}{2}+n)^2$，$n\in\mathbb{N}$

↳ 即为特征值，其对应的特征函数为 $\sin((\frac{1}{2}+n)x)$，$n\in\mathbb{N}$

(2) 令 $u(t,x)=T(t)X(x)$

从而由(1)知，$u(t,x)=\sum_{n=0}^{\infty} T_n(0)e^{-(\frac{1}{2}+n)^2 t}\sin((\frac{1}{2}+n)x)$

代回初始条件：$u(0,x)=\sin\frac{x}{2}=\sum_{n=1}^{\infty}T_n(0)\sin((\frac{1}{2}+n)x)$ 得 $T_n(0)=\begin{cases} 1 & , n=0 \\ 0 & , n\ge 1 \end{cases}$

故 $u(t,x)=e^{-\frac{1}{4}t}\sin\frac{x}{2}$

(3) $\partial_t u-\partial_{xx}u=f(t,x)$，$x\in(0,\pi)$，$t>0$

$f(t,x)=\sum_{n=0}^{\infty}f_n(t)\sin((\frac{1}{2}+n)x)$，其中 $f_n(t)=\frac{2}{\pi}\int_0^{\pi}f(t,y)\sin((\frac{1}{2}+n)y)\,dy$

故 $u(t,x)=e^{-\frac{1}{4}t}\sin\frac{x}{2}+\int_0^t ds\int_0^{\pi}f(t,y)\,G(t-s;x,y)\,dy$

其中 $G(t;x,y)=\frac{2}{\pi}\sum_{n=0}^{\infty}\sin((\frac{1}{2}+n)x)\sin((\frac{1}{2}+n)y)\,e^{-(\frac{1}{2}+n)^2 t}$

Ex 3. (1) 对于严格不等式：$\partial_t v-\Delta v>0$ 的情况. 设最小值在 $(t^*,x^*)\in\overline{\Omega}_T$ 达到

易证要么 $(t^*,x^*)\in\partial_p\Omega_T$，要么 $v(t^*,x^*)\ge 0$

事实上，若 $(t^*,x^*)\in\Omega_T$ 且 $v(t^*,x^*)<0$，则 $\partial_t v(t^*,x^*)\le 0$，$-\Delta v(t^*,x^*)\le 0$，矛盾

若 $(t^*,x^*)\in\partial_p\Omega_T$，由 $\frac{\partial v}{\partial n}(t^*,x^*)>0$ 知 $v(t^*,x^*)\ge 0$，即得结论.

进而令 $v_\varepsilon(t,x)=v(t,x)+\varepsilon t$，其 s.t. 严格不等式，从而 $v_\varepsilon(t,x)\ge 0$ on $\overline{\Omega}_T$

再令 $\varepsilon\to 0$ 即得结论.

(2) 考虑 $v_\varepsilon(t,x)=v(t,x)+\varepsilon(4t+x_1^2+x_2^2)$，$\varepsilon>0$

故 $\frac{\partial v_\varepsilon}{\partial n}(t,x)=\frac{\partial v}{\partial n}(t,x)+2\varepsilon>0$，易证 v_ε s.t. 上问其他条件

故 $v_\varepsilon(t,x)\ge 0$，即 $v(t,x)\ge -\varepsilon(4t+x_1^2+x_2^2)\ge -\varepsilon(4t+1)$ on $\overline{\Omega}_T$

令 $\varepsilon\to 0$ 即得结论.

(3) 令 $F := \sup_{\Omega_T} |f|$，$\Phi := \sup_{\Omega} |\varphi|$，$v(t,x) = tF + \Phi \pm u(t,x)$

则 $\partial_t v - \Delta v = F \pm f \geq 0$，$v(0,x) = \Phi \pm \varphi(x) \geq 0$，$\frac{\partial v}{\partial n} = \pm \frac{\partial u}{\partial n} \geq 0$

由上问知，$v(t,x) \geq 0$ on Ω_T，即得结论，$(C = T+1)$

Ex 4. (1) $\int_{\mathbb{R}} K(x,y)\,dy = \int_{-\infty}^{\infty} \frac{1}{\pi} \frac{x_1}{(x_2-y)^2 + x_1^2}\,dy = \frac{x_1}{\pi} \cdot \frac{1}{x_1} \arctan\left(\frac{y}{x_1}\right)\Big|_{-\infty}^{\infty} = 1.$

$$\lim_{h\to 0^+} \int_{z-\varepsilon}^{z+\varepsilon} K((h,z),y)\,dy = \lim_{h\to 0^+} \int_{z-\varepsilon}^{z+\varepsilon} \frac{1}{\pi} \frac{h}{(z-y)^2+h^2}\,dy$$

$$= \lim_{h\to 0^+} \frac{1}{\pi} \arctan\left(\frac{y}{h}\right)\Big|_{-\varepsilon}^{\varepsilon} = \lim_{h\to 0^+} \frac{1}{\pi}\left[\arctan\left(\frac{\varepsilon}{h}\right) - \arctan\left(-\frac{\varepsilon}{h}\right)\right] = 1.$$

(2) 对 $\forall \varepsilon > 0$，$\exists \delta > 0$，有 $|y-z| < \delta$ 时，$|g(y) - g(z)| < \varepsilon$ （$\because g$ 一致连续）

故对 $\forall z \in \mathbb{R}$，
$$\begin{aligned}
|g_h(z) - g(z)| &\leq \int_{-\infty}^{z-\delta} K((h,z),y)|g(y)-g(z)|\,dy \\
&\quad + \int_{z-\delta}^{z+\delta} K((h,z),y)|g(y)-g(z)|\,dy \\
&\quad + \int_{z+\delta}^{\infty} K((h,z),y)|g(y)-g(z)|\,dy \\
&\leq 2\int_{\mathbb{R}\setminus(z-\delta,z+\delta)} K((h,z),y)\,dy \\
&\quad + \varepsilon \int_{z-\delta}^{z+\delta} K((h,z),y)\,dy \\
&\leq 2\left(1 - \frac{2}{\pi}\arctan\left(\frac{\varepsilon}{h}\right)\right) + \varepsilon
\end{aligned}$$

故可取合适的 h，使上式 $< \varepsilon$，从而 g_h 一致收敛到 g